"Geometric" stability analysis (Lyapunov)

Examine behavior of $\frac{d}{dt}\|\underline{x}(t)\|$ via different metrics

[Initially homogeneous / "internal" stability]

for example: $\frac{d}{dt}\|\underline{x}(t)\|_2^2 = \underline{x}^T(t)\left[\underline{A}^T(t)+\underline{A}(t)\right]\underline{x}(t)$

$= \underline{x}^T(t)\,\underline{M}(t)\,\underline{x}(t)$

which is a quadratic form.

what can we say generally about such forms, and in particular when can they be guaranteed to be negative?

Quadratic form: any expression like

$$\underline{x}^T\underline{M}\,\underline{x}$$

where $\underline{M}$ is square, and same size as $\underline{x}$

This defines a scalar, equivalent to

$$\underline{x}\cdot(\underline{M}\,\underline{x})$$

i.e. the dot product of $\underline{x}$ with $\underline{M}\underline{x}$

Note any square matrix $\underline{M}$ can be written as

$$\underline{M} = \underline{M}_{sym} + \underline{M}_{skew}$$

$$\underline{M}_{sym}^T = \underline{M}_{sym} \quad \text{(symmetric)}$$

$$\underline{M}_{skew}^T = -\underline{M}_{skew}$$

in fact

$$\underline{M}_{sym} = \tfrac{1}{2}(\underline{M}+\underline{M}^T)$$

$$\underline{M}_{skew} = \tfrac{1}{2}(\underline{M}-\underline{M}^T)$$

Then $\underline{x}^T\underline{M}\,\underline{x} = \underline{x}^T(\underline{M}_{sym}+\underline{M}_{skew})\underline{x}$

$= \underline{x}^T\underline{M}_{sym}\,\underline{x} + \underline{x}^T\underline{M}_{skew}\,\underline{x}$

Note

$$\underline{x}^T\underline{M}_{skew}\,\underline{x} = \left(\underline{x}^T\underline{M}_{skew}\,\underline{x}\right)^T$$

$$= \underline{x}^T\underline{M}_{skew}^T\,\underline{x}$$

so $\underline{x}^T\underline{M}_{skew}\,\underline{x} = -\left(\underline{x}^T\underline{M}_{skew}\,\underline{x}\right)$

$$\Rightarrow \underline{x}^T\underline{M}_{skew}\,\underline{x} = 0$$

Hence only the symmetric part of M contributes to the quadratic form.

=> generally assume the matrix in a quadratic form is symmetric

Symmetric matrices have real eigenvalues which leads to the Rayleigh-Ritz inequality

$$\lambda_{min}(M)\|\underline{x}\|_2^2 \leq \underline{x}^T M \underline{x} \leq \lambda_{max}(M)\|\underline{x}\|_2^2$$

Above we showed

$$\frac{d}{dt}\|\underline{x}\|_2^2 = \underline{x}^T M(t)\,\underline{x}$$

with $M(t) = A^T(t) + A(t)$

$$\Rightarrow \quad \frac{d}{dt}\|\underline{x}\|_2^2 \leq \alpha(t)\|\underline{x}\|_2^2$$

$$\alpha(t) = \max_k \lambda_k\left[A^T(t) + A(t)\right]$$

$$\Rightarrow \text{if } \max_t \alpha(t) \leq 0$$

$$\Rightarrow \text{system is stable i.s.L.}$$

Thm:

if $\max_t \lambda_{max}\left[A^T(t) + A(t)\right] \leq 0$

then system is stable i.s.L.

Careful, this does not mean that $\alpha(t) > 0 \ \forall t$ implies instability! It might, or might not => requires further investigation.

More careful / formal def'ns:

Stable (i.s.L.): if $\exists\, \gamma(t_0) < \infty$ so that

$$\|\underline{x}(t)\| \leq \gamma(t_0)\|\underline{x}_0\| \quad \forall t \geq t_0$$

If we can find a single γ, indep. of t_0, that works for all t_0, the stability is said to be "uniform".

Example where γ depends on t_0

$$\dot{x}(t) = a(t)\,x(t)$$

$$a(t) = 4t\sin 2t - 2t$$

with $t_0 = 2K\pi$ $(K = 0, 1, 2, \ldots)$

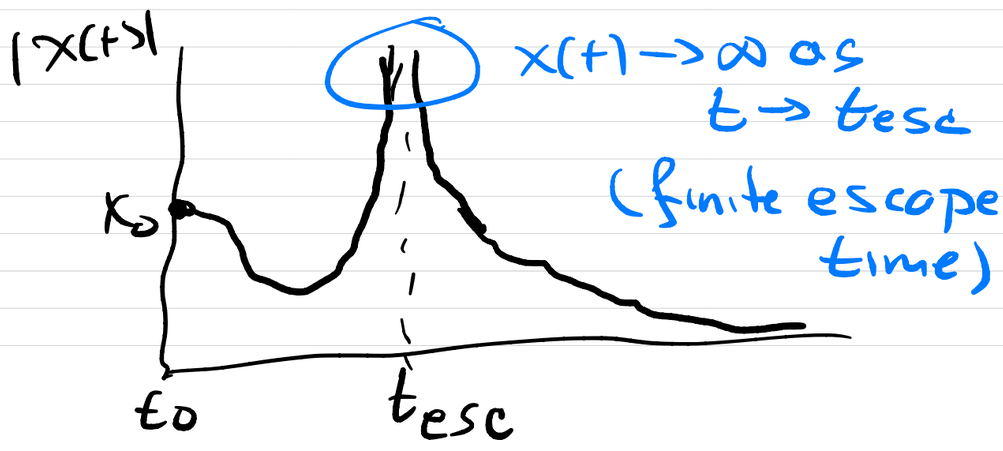
sol'n for $x(t)$ reaches a max at $t = t_0 + \pi$, for which

$$\frac{|x(t_0+\pi)|}{|x_0|} = \underbrace{\exp\left[(2t_0+\pi)(4-\pi)\right]}_{\gamma(t_0)}$$

Asymptotic Stability

1.) stable (s.L, and)

2.) $\|x(t)\| \to 0$ as $t \to \infty$



2) can be expressed as:

for any $\varepsilon > 0$, $\exists\, T(\varepsilon, t_0)$ s.t. that $\|x(t)\| \le \varepsilon \|x_0\|$

$\forall\, t \ge t_0 + T$.

If a single T can be found that works for every t_0, the system is said to be uniformly asymp. stab. (UAS)

Note: T will still depend on ε!

= Example: linear system w/ fund'l sol'n $\varphi(t) = \frac{1}{1+t}$ and STM

$$\Phi(t, t_0) = \frac{1+t_0}{1+t}$$

so $x(t) = \left[\frac{1+t_0}{1+t}\right] x_0$

system is A.S.
is it UAS?

$$\frac{|x(t)|}{|x_0|} \leq \frac{1+t_0}{1+t}$$

Suppose we take $\varepsilon = 1/2$

Then we need to find t such that

$$\frac{1}{2} \geq \frac{1+t_0}{1+t}$$

if $t_0 = 0 \Rightarrow t \geq 1 = 1 + t_0$

$$\Rightarrow T(1/2, 0) = 1$$

if $t_0 = 10$

$$\frac{|H0|}{1+t} = \frac{1+t_0}{1+t} \leq \frac{1}{2}$$

$$\Rightarrow t \geq 21 = t_0 + 11$$

so $T(1/2, 10) = 11$

Not uniformly Asymp. stable

Generally here T must satisfy (T depends on both ε and t_0!)

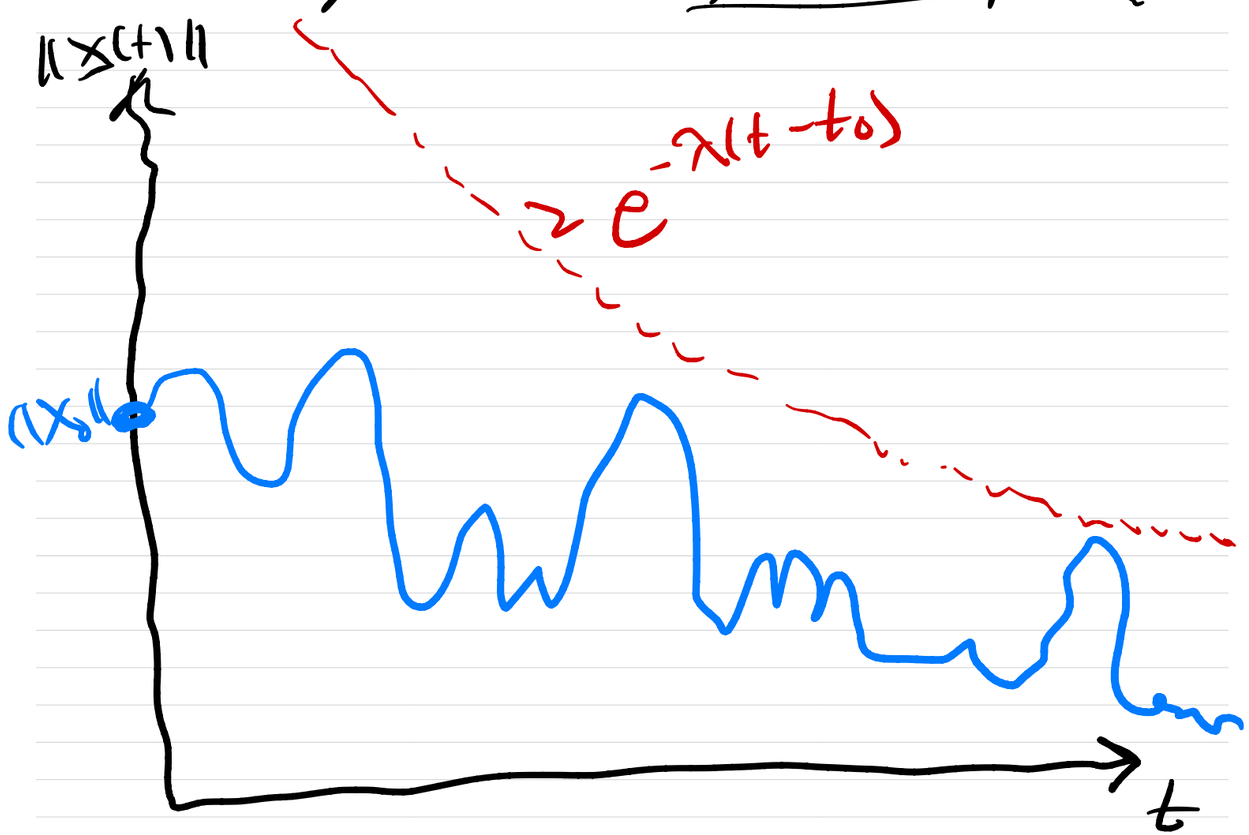
$$\frac{1+t_0}{1+t_0+T} \leq \varepsilon$$

Exp. Stab:

if $\exists \gamma(t_0) < \infty$ and $\lambda(t_0) > 0$ such that

$$\|x(t)\| \leq \gamma e^{-\lambda(t-t_0)} \|x_0\| \quad \forall t \geq t_0$$

Then system is exponentially stable



ES is uniform if both γ and λ are constants, indep. of t_0.

Note: ES → AS

But reverse is not generally true.

However, for Linear systems (Thm)
it is true that UAS → UES.

Proof:

Recall: $\underline{x}(t) = \Phi(t, t_0)\,\underline{x}_0$

$\Rightarrow \|\underline{x}(t)\| \le \|\Phi(t, t_0)\|\,\|\underline{x}_0\|$

UAS $\Rightarrow$ $\exists T(\varepsilon) < \infty$ so that

$\|\Phi(t_0+T, t_0)\| \le \varepsilon$ for any t_0

= Chop time into intevals of length t $[t_0, t_0+T]$, $[t_0+T, t_0+2T)$ etc.

1.) for $t \in [t_0, t_0+T]$

$\|\underline{x}(t)\| \le \gamma \|\underline{x}_0\|$

2.) $t \in [t_0+T, t_0+2T]$

$\underline{x}(t) = \Phi(t, t_0)\,\underline{x}_0$

$= \Phi(t, t_0+T)\,\Phi(t_0+T, t_0)\,\underline{x}_0$

$\|\underline{x}(t)\| \le \underbrace{\|\Phi(t, t_0+T)\|}_{\gamma}\,\underbrace{\|\Phi(t_0+T, t_0)\|}_{\varepsilon}\,\|\underline{x}_0\|$

$\|\underline{x}(t)\| \le \gamma\varepsilon\|\underline{x}_0\|$

3.) $t \in [t_0+2T, t_0+3T]$

$\underline{x}(t) = \Phi(t, t_0+2T)\,\Phi(t_0+2T, t_0+T)\,\Phi(t_0+T, t_0)\,\underline{x}_0$

$\|\underline{x}(t)\| \le \gamma\varepsilon \cdot \varepsilon \cdot \|\underline{x}_0\|$

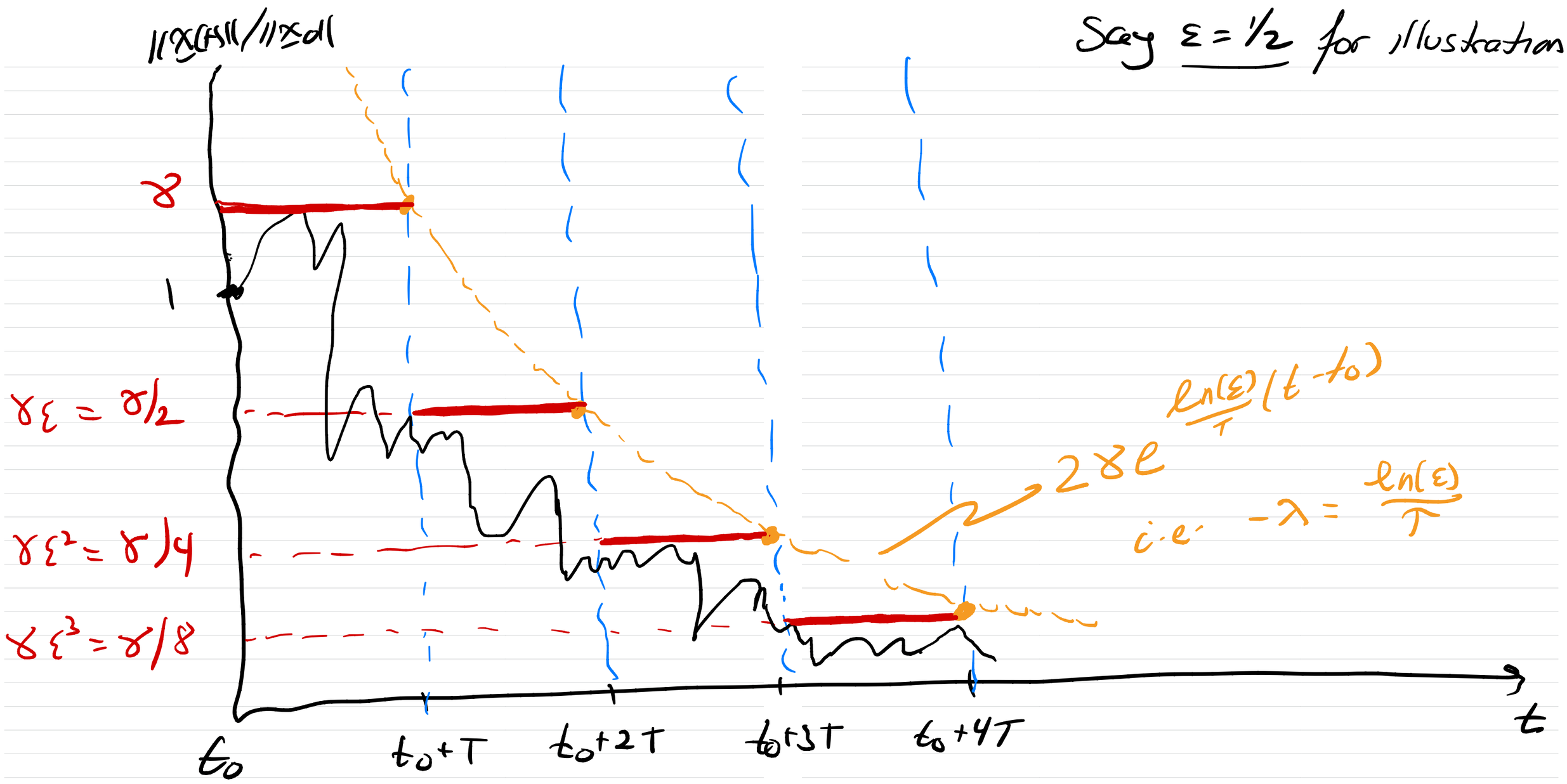
$\le \gamma\varepsilon^2\|\underline{x}_0\|$

4.) for $t \in [t_0+KT, t_0+(K+1)T]$

$\|\underline{x}(t)\| \le \gamma\varepsilon^K\|\underline{x}_0\|$

$\to 0$ as $K \to \infty$ if $\varepsilon < 1$

Say $\varepsilon = 1/2$ for illustration



Thus UAS $\Rightarrow$ $\|\underline{x}(t)\| \le \gamma_1 e^{-\lambda(t-t_0)} \|\underline{x}_0\| \Rightarrow$ UES

(equal contraction - ε - on equal time intervals - T)

with $\lambda = -\frac{\ln(\varepsilon)}{T}$

and $\varepsilon, T(\varepsilon)$ describe the asymp. Stab. properties of system